

Probabilistic Coherence as a Test of Clinical LLM Reasoning

Abstract

Bayesian belief revision is order invariant as a theorem, since $P(d \mid e_1 \dots e_n)$ is a function of the evidence set and not of the sequence in which it arrives. This gives a correctness criterion for clinical language models that needs no annotator, no judge model and no independence assumption, because the target value is known to be zero exactly. We evaluate eleven open weight models from 4B to 32B parameters, spanning general, medically tuned and reasoning distilled families, on 1.12 million elicited responses over 1,956 DDXPlus patients and 49 pathologies. Every divergence is reported against three anchors measured on the same model, namely its test retest floor, its between patient ceiling, and a corpus level generator noise floor of 0.0490 established from exact replicate cases in DDXPlus itself. Three findings follow. The anchors reverse the direction of the headline result, since raw permutation divergence rises with model scale while the anchored order effect does not. Belief updates do not track the evidence, with direction agreement against an assumption free empirical oracle spanning 0.454 to 0.549 across all eleven models. Most consequentially, every model names the correct diagnosis two to five times more often than its own stated posterior ranks it first, and agrees with the top entry of its own distribution on only 3 to 12 percent of items. The deficit is therefore in the probabilistic interface rather than in the clinical knowledge, which motivates ELR Fusion, a symbolic combination layer attaining an exactly zero order effect at zero marginal query cost in all 132 configurations tested.

1. Methodology

1.1 Coherence axioms

The instrument tests four properties of belief revision, each with an exact target and none requiring human annotation. A1 and A4 are label free; A2 and A3 need ground truth, and the choice of reference is the main methodological commitment of the study.

Axiom	Property	Target
A1	Order invariance, $P(d \mid e_1 \dots e_n)$ depends on the evidence set and not its order	0
A2	Update fidelity, adding a finding moves belief as the corpus says it should	slope 1
A3	Redundancy insensitivity, certified uninformative evidence must not move belief	0
A4	Positional anchoring, no position in the sequence carries special weight	0

1.2 Data, models and the absent knowledge base

The corpus is DDXPlus [1], comprising 1,292,579 synthetic patients over 49 pathologies and 223 findings. A data audit established two facts that constrain the design. The released knowledge base declares 888 pathology by evidence slots and supplies conditional probabilities for none of them, so exact posteriors for arbitrary evidence subsets cannot be read off it. Reconstructing those tables from corpus frequencies and recombining them under conditional independence reproduces the shipped differentials at mean Jensen Shannon divergence 0.546, roughly eleven times the generator noise floor, and was therefore rejected. The posterior is instead conditioned directly on all 1,292,579 released patients with Wilson intervals and a minimum support of 500 cases, which makes A2 and A3 absolute while making no independence assumption.

The audit also found the ground truth differential to be stochastic. Across 21,004 replicate pairs, that is patients identical in evidence multiset, age, sex, initial evidence and true pathology, the shipped differential differs for 25.8 percent of pairs at mean divergence 0.0490. Every DDXPlus differential matching metric therefore has a ceiling below 1.0, which applies to any work on this corpus. Eleven models were evaluated: Qwen3 at 4B, 8B and 32B in thinking and nonthinking configurations, MedGemma 4B, MedGemma 27B, Med42 8B, R1 Distill 32B and GPT OSS 20B.

1.3 Three anchors and a competence gate

A raw divergence is uninterpretable alone, since a model that resamples noisily appears incoherent regardless of ordering. Each model is measured against three references computed on that same model and the same items: a test retest floor from resampling the identical prompt, a between patient ceiling from unrelated patients, and the corpus generator floor of 0.0490. The anchored order effect is the gap from the floor as a fraction of the floor to ceiling distance.

A competence gate precedes all coherence reporting, because schema validity constrains the shape of a response and not its content, and a model returning a constant scores a near zero order effect that reads as excellent coherence. During calibration GPT OSS 20B under a JSON grammar returned an identical minimal object for essentially every case at 1.000 schema validity and 1.000 informative rate, scoring 0.039 top 1 accuracy against a chance rate of 0.020. The cause is the harmony response format, which requires a channel token at position zero that a JSON grammar forbids. Decoded freely and parsed from the final channel, the same model on the same items reaches 0.653, a change of 16.6 times from one decoding flag. All reported results use the corrected path.

1.4 ELR Fusion

Log odds accumulate over an evidence set, so the result is order invariant by construction: $\text{logodds}(d \mid E) = \text{logodds}(d) + \tau \sum_i \log \text{LR}(e_i \mid d)$. The likelihood ratio table is elicited once per finding for the whole corpus, so a new case costs zero additional model queries. Because the sum ranges over an unordered set, and scaling an order invariant quantity leaves it order invariant, the guarantee holds for every value of τ . That coefficient is the log odds form of temperature scaling [6] applied to the evidence term so the prior is untouched, correcting the overcounting that arises because correlated findings are summed as though independent. It is fitted by cross fitting over two folds, so the coefficient applied to a case is never fitted on that case.

2. Results

2.1 A1, the anchors reverse the finding

Permutation divergence rises with scale across the Qwen3 ladder, from 0.128 at 4B to 0.291 at 32B, which read alone would indicate that larger models are less coherent. The test retest floor rises faster over the same ladder, from 0.072 to 0.256, so the anchored effect falls from 0.215 to 0.127. Reporting the raw number alone inverts the conclusion, which is the strongest argument for the three anchor design and the reason prior order effect figures are not directly comparable to these.

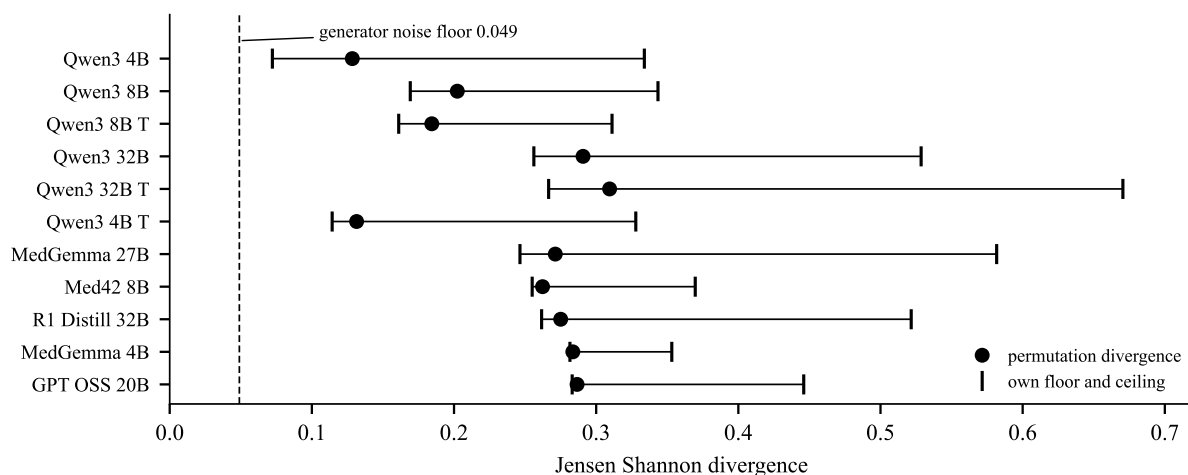


Figure 1: Order invariance against three anchors. For each model the horizontal line spans its own test retest floor to its own between patient ceiling, and the filled circle is the observed permutation divergence. The order effect is the gap from the floor, not the observed value.

The effect is real but not universal. Eight of eleven models show an order effect whose bootstrap interval excludes zero. For MedGemma 4B, Med42 8B and GPT OSS 20B the interval covers zero, and in each case the floor has risen close to the ceiling, which is a collapsing denominator rather than a coherence success. The top 1 flip rate is high throughout, from 0.699 to 0.994, so reordering the same findings changes the leading diagnosis for most patients even where distributional movement is modest.

2.2 A2, A3 and A4

Axiom	Quantity	Range over eleven models	Ideal
A2	Slope β against the empirical oracle	0.0009 to 0.0576	1.0
A2	Coefficient of determination r^2	at most 0.0207	1.0
A2	Direction agreement	0.454 to 0.549	1.0
A3	Mean divergence from uninformative evidence	0.0759 to 0.2687	0
A3	Cases moved beyond the model's own floor	15.8 to 44.9 percent	0
A3	Ninetieth percentile of that divergence	up to 0.928	0
A4	Models with a significant positional slope	7 of 11	0 of 11

A2 is the strongest negative result. Whether a model's belief in a diagnosis rises or falls when a finding is added is uncorrelated with whether the corpus says it should, and direction agreement straddles chance for every model. Six of eleven have a slope interval excluding zero, so the effect is real, but at β below 0.058 against an ideal of 1.0 it is swamped. Unlike A1 this is a failure of direction measured against ground truth, with no independence assumption available to attack.

A3 requires care. Evidence the corpus certifies as uninformative moves the posterior beyond the model's own resampling noise for 15.8 to 44.9 percent of cases and flips the leading diagnosis for 26.5 to 69.8 percent. The mean spurious divergence, however, sits below the model's own retest floor for ten of eleven models. The distribution is heavy tailed rather than uniformly shifted, so the correct claim is that irrelevant findings move belief substantially for a large minority of cases, not that they do so on average.

For A4 the directional reading matters. A divergence based position measure is near zero throughout, between -0.004 and 0.014 , but Jensen Shannon divergence is symmetric and cancels a consistent directional shift. Regressing the target log odds on position instead, seven of eleven models show a slope whose bootstrap interval excludes zero. Six are primacy, meaning the decisive finding carries more weight when presented first, and they comprise every Qwen3 configuration together with R1 Distill 32B. MedGemma 27B alone reverses to recency. A symmetric divergence measure is blind to the effect the regression detects, which is worth noting for anyone measuring position bias this way.

2.3 The answer and belief gap

Each model was asked, on the same item and with the same prompt content, both to name one diagnosis and to score all 49 candidates for plausibility. The two tasks share an item identifier, so the comparison is exactly paired. Every model is markedly more accurate when naming a diagnosis than when ranking its own distribution. Self agreement below is the share of items on which the named diagnosis equals the top entry of that model's own posterior.

Model	Paired items	Named	Posterior	Ratio	Self agreement	Median rank
Med42 8B	35,208	0.309	0.079	3.89	0.121	3
Qwen3 8B	35,208	0.281	0.109	2.58	0.085	3
Qwen3 8B think	35,208	0.252	0.097	2.58	0.067	3
Qwen3 4B	35,205	0.267	0.059	4.51	0.042	4
Qwen3 4B think	32,696	0.235	0.049	4.81	0.019	3

A McNemar test on Med42 8B gives $\chi^2 = 7,131$ over 8,633 versus 543 discordant pairs. Three confounds were excluded. Uniform posteriors cannot explain the effect, since Med42 8B emits a uniform vector on 0.3 percent of items and shows the largest gap. Ties at the top cannot explain it, since restricting to strictly unimodal posteriors leaves accuracy at 0.081 against 0.309 named. Ordering cannot explain it, since the comparison is paired within item. The median rank of the model's own named diagnosis inside its own distribution is 3 or 4 of 49, against a chance value of 25. The posterior therefore encodes the answer and declines to rank it first. This is not a knowledge

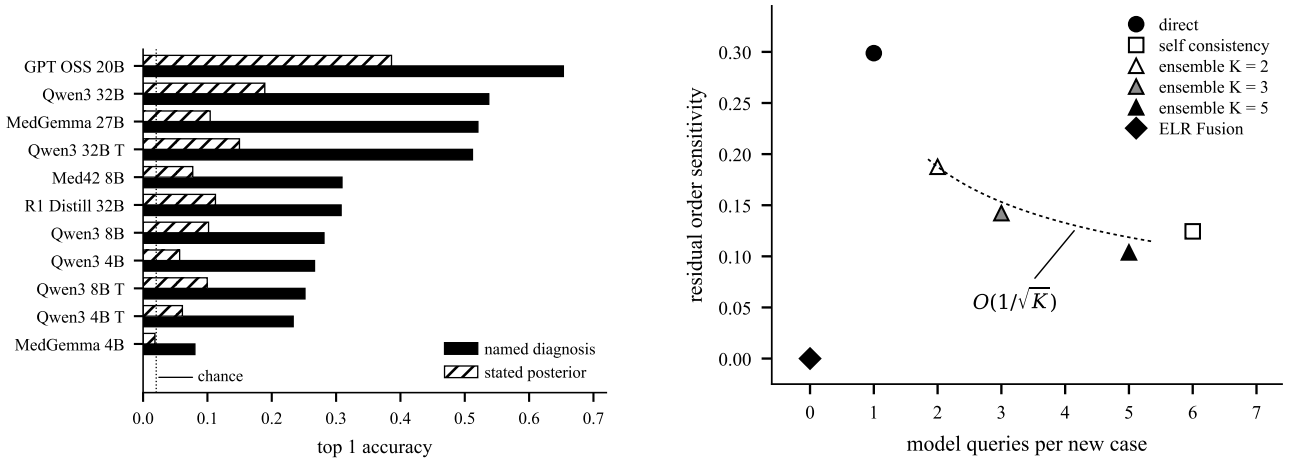


Figure 2: Left, the answer and belief gap across all eleven models. Solid bars are top 1 accuracy when the model names one diagnosis; hatched bars are the top ranked entry of the same model’s own stated posterior on the same items. Right, residual order sensitivity against query cost for Qwen3 32B. Permutation ensembling decays as $O(1/\sqrt{K})$ and never reaches zero, while ELR Fusion is exactly zero at zero marginal cost.

failure and not sampling noise. It is a failure to express what the model knows as a coherent distribution, and it explains the A2 null, since a model whose probabilistic interface is broken cannot move that interface in the direction the evidence requires.

2.4 Remedies

Paired over the cases every method produced, for Qwen3 32B:

Method	Order residual	Queries per case	Top 1	Top 5	Entropy	Exact
Direct	0.2988	1	0.1892	0.3006	3.94	no
Self consistency	0.1244	6	0.2265	0.4346	4.50	no
Permutation ensemble K5	0.1038	5	0.2377	0.4443	4.53	approximate
ELR Fusion	0.0000	0	0.1334	0.2480	0.86	yes
ELR Fusion calibrated	0.0000	0	0.0997	0.2393	5.25	yes

ELR Fusion is the only method reaching an exactly zero order effect, at zero marginal query cost, because the weight table is elicited once for the whole corpus rather than once per case. The result generalises: residual order sensitivity is exactly zero for all eleven models at all twelve values of τ , that is 132 of 132 configurations. Permutation ensembling buys its residual with queries, decays only as $O(1/\sqrt{K})$, and no value of K reaches zero.

The honest cost is accuracy, and it is model dependent rather than fixed. Across the eleven models ELR Fusion retains between 0.30 and 1.06 of direct elicitation accuracy, median 0.71, and for two models it is more accurate than direct elicitation. The shrinkage coefficient is likewise a frontier rather than a fitted constant, since the accuracy optimal τ reaches at least 2.0 for eight of eleven models while the optimum for the likelihood of the truth falls at or below 0.10 for ten of eleven, a difference of more than an order of magnitude in the same model. Order invariance is exact at every point on that frontier.

3. Conclusion

Three claims are supported. First, order effects in clinical language models cannot be interpreted without anchors measured on the same model, because the test retest floor rises with scale faster than the effect does and the raw number therefore inverts the comparison. Second, the failure these models exhibit is not primarily a reasoning failure. They identify the correct diagnosis at up to 32 times the chance rate while ranking that same diagnosis third or fourth of 49 in their own stated distribution, and they agree with themselves on fewer than one item in eight. The deficit lies in the probabilistic interface. Third, because the deficit is one of representation rather than knowledge, moving the probability arithmetic into a symbolic layer is the structurally indicated fix, and ELR Fusion attains an exactly zero order effect at zero marginal query cost in every configuration tested, at an accuracy cost reported in full rather than concealed.

Two limitations are stated plainly. DDXPlus is synthetic, which is what makes an exact reference possible for A2 and A3, but an external validity arm on real clinical narratives is required before these magnitudes are claimed for deployed settings. The elicited likelihood ratios, rather than the fusion rule, set the accuracy ceiling of the method, and improving that elicitation is where the remaining accuracy lies.

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